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Resource-Constrained CNNs for Edge Image Classification

Waste Material Classification of RealWaste Images

EN3150 Assignment 03

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Executive Summary

Deploying vision-based automated material sorting on autonomous recycling collection nodes requires balancing high classification accuracy against strict hardware limitations. Embedded microcontrollers and edge perception sensors often operate with under 512 KB of SRAM, 1–2 MB of flash memory, and battery-constrained thermal power envelopes.

In this investigation, Team Outliers investigates the design, parameter optimization, arithmetic complexity, and hardware profiling of edge-oriented vision models on the 9-class RealWaste dataset downscaled to 64×64 resolution. We design a conventional 3-stage convolutional network (**Model A**, 102,313 parameters) and architect a highly compact depthwise-separable network (**Model B**, 22,297 parameters) engineered strictly beneath the 100,000 parameter budget. Furthermore, we benchmark these custom models against fine-tuned state-of-the-art mobile backbones (**MobileNetV2** and **ShuffleNetV2 1.0**×).

Our empirical results demonstrate that Model B achieves **69.85% top-1 test accuracy**, surpassing Model A (67.32%) while reducing parameter count by **78.2%** and multiply-accumulate operations (MACs) by **85.0%** (from 41.30M to 6.19M). Model B occupies only **0.10 MB (98 KB)** of serialized memory, making it uniquely suited for on-chip deployment on resource-constrained microcontrollers like the STM32H7 or ESP32-S3. Conversely, while MobileNetV2 and ShuffleNetV2 achieve higher classification accuracy (88.22% and 88.64%), their substantial flash footprints (> 4.9 MB) and memory traffic render them prohibitive for microcontrollers, establishing an empirical Pareto frontier for smart recycling edge infrastructure.

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1 Introduction and Edge Computing Objectives

Automated municipal and industrial solid-waste sorting is a vital prerequisite for modern resource recovery and circular economy initiatives. Conventional sorting facilities rely on centralized sensor arrays coupled to remote server clusters. However, transmitting uncompressed high-resolution video streams across wireless links introduces latency bottlenecks, requires sustained network availability, and exposes municipal data streams to network congestion.

Edge intelligence provides an alternative paradigm: executing neural inference directly at the optical sensor node. In such setups, camera modules embedded into sorting chutes or autonomous collection bins categorize materials immediately upon image capture. The primary challenge stems from the operational envelope of edge silicon:

- **Strict Memory Boundaries:** Microcontrollers (e.g., ARM Cortex-M4/M7, ESP32-S3) possess small on-chip static RAM (often 256 KB to 512 KB) and limited non-volatile NOR Flash (typically 1 MB to 2 MB). Large neural weights cannot be buffered or executed without external memory chips, which dramatically increases unit cost and power draw.
- **Arithmetic and Energy Budgets:** Low-power edge processors lack dedicated desktop GPUs. Every multiply-accumulate (MAC) operation consumes valuable clock cycles and battery energy.
- **Visual Ambiguity at Low Spatial Resolution:** To conform to buffer limits, edge frames must be downscaled (here, 64×64 pixels), which compresses fine textures and intensifies inter-class ambiguity between visually similar waste streams.

This investigation explores these trade-offs by evaluating custom convolutional architectures against modern mobile transfer learning backbones, providing empirical guidance for edge machine vision design.

2 RealWaste Dataset Analysis and Pipeline Engineering

2.1 Dataset Formulation and Environmental Realism

We utilize the RealWaste benchmark from the UCI Machine Learning Repository [1]. Unlike pristine laboratory datasets, RealWaste depicts municipal waste items under uncontrolled lighting, deformation, partial occlusion, and complex backgrounds.

The dataset spans 9 distinct classes: *Cardboard*, *Food Organics*, *Glass*, *Metal*, *Miscellaneous Trash*, *Paper*, *Plastic*, *Textile Trash*, and *Vegetation*. The original images are downscaled to $64 \times 64 \times 3$ RGB tensors. This reduces raw frame size from 147 KB (224×224) to approximately 12 KB (64×64) — a 91.8% reduction in sensor buffer requirements, making real-time streaming viable on low-speed SPI/DVP camera interfaces.

2.2 Deterministic Stratified Partitioning

To prevent data leakage and guarantee reproducible evaluation, the 4,752 images are partitioned into training, validation, and test sets using a fixed stratified split of 70% (3,326 images), 15% (713 images), and 15% (713 images) using random seed 42.

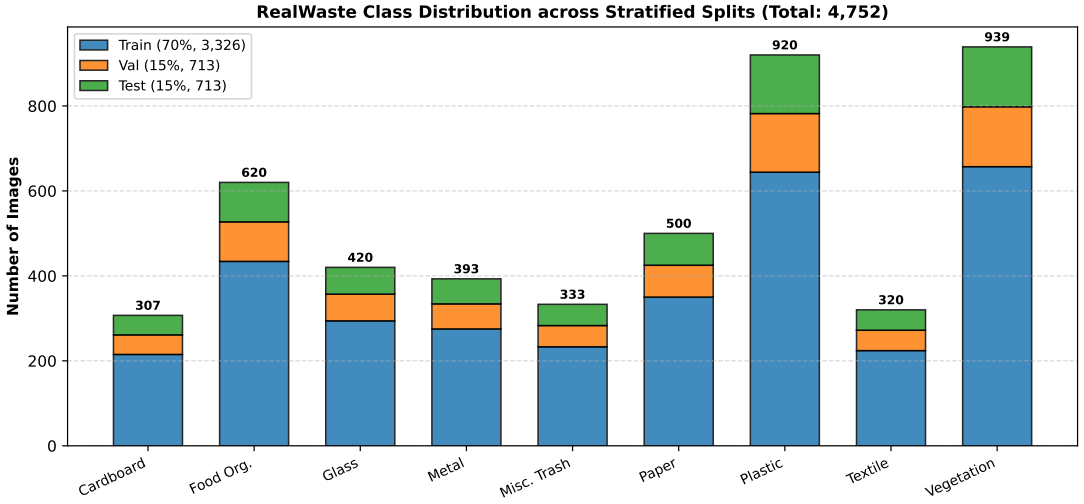


Figure 1: Class distribution across the 70% training, 15% validation, and 15% test splits of RealWaste (4,752 images). Proportions are strictly preserved across all three partitions.

Table 1 summarizes the split statistics. The natural class imbalance — where *Plastic* and *Vegetation* represent large shares while *Textile Trash* is less frequent — is accurately mirrored across all subsets.

Table 1: Deterministic stratified dataset partition details.

Split Subset	Sample Count	Percentage	Batch Count ($B = 64$)	Usage Role
Training Set	3,326	70.0%	52	Backpropagation & Weight Updates
Validation Set	713	15.0%	12	Checkpointing & Early Stopping
Test Set	713	15.0%	12	Final Unseen Metric Reporting
Full Dataset	**4,752**	**100.0%**	**76**	—

2.3 Preprocessing and Augmentation Strategy

Inputs are normalized using channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Training tensors undergo random horizontal flips ($p = 0.5$), random vertical flips ($p = 0.2$), and slight affine rotations ($\pm 10°$) to mitigate overfitting on smaller sample classes without corrupting spatial geometry.

3 Architectural Design: Baseline vs. Depthwise Separable CNN

3.1 Mathematical Foundations of Lightweight Convolutions

Standard 2D convolution applies a 4D kernel tensor $\mathbf{W} \in \mathbb{R}^{C_{out} \times C_{in} \times K \times K}$ simultaneously across all spatial and channel dimensions. For an input feature map $\mathbf{X} \in \mathbb{R}^{C_{in} \times H \times W}$, computing an output activation at spatial coordinate (i, j) requires:

$$\mathbf{Y}_{k,i,j} = \sum_{c=1}^{C_{in}} \sum_{m=-k_h}^{k_h} \sum_{n=-k_w}^{k_w} \mathbf{W}_{k,c,m,n} \, \mathbf{X}_{c,i+m,j+n} + \mathbf{b}_k \tag{1}$$

The computational cost in MAC operations is:

$$\text{MACs}_{\text{standard}} = H_{out} \times W_{out} \times C_{in} \times C_{out} \times K^2 \tag{2}$$

Depthwise Separable Convolutions (pioneered by Howard et al. [2]) factorize this operation into two disjoint phases:

1. **Depthwise Spatial Convolution:** A single $K \times K$ filter is applied per input channel independently ($G = C_{in}$ groups):

$$\text{MACs}_{\text{depthwise}} = H_{out} \times W_{out} \times C_{in} \times K^2 \tag{3}$$

2. **Pointwise Channel Projection:** A 1×1 linear convolution combines channel representations across the depth:

$$\text{MACs}_{\text{pointwise}} = H_{out} \times W_{out} \times C_{in} \times C_{out} \times 1^2 \tag{4}$$

The ratio of arithmetic operations is given analytically by:

$$\frac{\text{MACs}_{\text{depthwise}} + \text{MACs}_{\text{pointwise}}}{\text{MACs}_{\text{standard}}} = \frac{K^2 \cdot C_{in} + C_{in} \cdot C_{out}}{K^2 \cdot C_{in} \cdot C_{out}} = \frac{1}{C_{out}} + \frac{1}{K^2} \tag{5}$$

For 3×3 kernels ($K = 3$), this yields an asymptotic $8\times$ to $9\times$ reduction in computational burden with negligible degradation in representation capability.

3.2 Baseline Architecture: Model A

Model A serves as a standard convolutional reference. It consists of three cascaded convolutional stages with filter counts of 32, 64, and 128:

- **Stage 1:** Conv2d($3 \to 32, K = 3, P = 1$) $\to$ BatchNorm $\to$ ReLU $\to$ MaxPool2d(2)
- **Stage 2:** Conv2d($32 \to 64, K = 3, P = 1$) $\to$ BatchNorm $\to$ ReLU $\to$ MaxPool2d(2)
- **Stage 3:** Conv2d($64 \to 128, K = 3, P = 1$) $\to$ BatchNorm $\to$ ReLU $\to$ MaxPool2d(2)
- **Classifier:** Global Average Pooling (GAP) $\to$ Dropout(0.3) $\to$ Linear($128 \to 9$).

Parameter Calculation for Model A:

- Conv1: $(3 \times 3 \times 3) \times 32 + 32 = 896$ params; BN1: $2 \times 32 = 64$ params.
- Conv2: $(3 \times 3 \times 32) \times 64 + 64 = 18,496$ params; BN2: $2 \times 64 = 128$ params.
- Conv3: $(3 \times 3 \times 64) \times 128 + 128 = 73,856$ params; BN3: $2 \times 128 = 256$ params.
- Linear: $128 \times 9 + 9 = 1,161$ params.
- **Total Model A Parameters:** $896 + 64 + 18,496 + 128 + 73,856 + 256 + 1,161 = \mathbf{102,313}$.

3.3 Lightweight Architecture: Model B ($< 100,000$ Parameters)

Model B is purpose-built to comply strictly with the assignment requirement of staying under 100,000 trainable parameters. It incorporates:

1. A compact standard convolution stem ($3 \to 24$ channels) to encode low-level RGB edges.
2. Three Depthwise Separable Blocks ($24 \to 48$, $48 \to 96$, $96 \to 128$).
3. ReLU6 non-linearities: $\text{ReLU6}(x) = \min(\max(0, x), 6)$. Bounding activations prevents numerical dynamic-range overflow and facilitates quantization into 8-bit fixed-point integers (INT8).
4. Global Average Pooling eliminating heavy dense layers before the 9-class linear projection.

Table 2: Layer-by-layer architectural comparison between Model A and Model B.

Model A (Standard CNN)			**Model B (Depthwise Separable CNN)**		
Layer Type	**Output Shape**	**Params**	**Layer Type**	**Output Shape**	**Params**
Conv2d + BN	$32 \times 64 \times 64$	960	Stem Conv2d + BN	$24 \times 64 \times 64$	696
MaxPool2d	$32 \times 32 \times 32$	0	MaxPool2d	$24 \times 32 \times 32$	0
Conv2d + BN	$64 \times 32 \times 32$	18,624	DS-Block 1 (DW+PW)	$48 \times 32 \times 32$	1,536
MaxPool2d	$64 \times 16 \times 16$	0	MaxPool2d	$48 \times 16 \times 16$	0
Conv2d + BN	$128 \times 16 \times 16$	74,112	DS-Block 2 (DW+PW)	$96 \times 16 \times 16$	5,280
MaxPool2d	$128 \times 8 \times 8$	0	MaxPool2d	$96 \times 8 \times 8$	0
—	—	—	DS-Block 3 (DW+PW)	$128 \times 8 \times 8$	13,624
GAP + Dense	9	1,161	GAP + Dense	9	1,161
Total	**Params**	**102,313**	**Total**	**Params**	**22,297**

Model B utilizes only **22,297 trainable parameters**, which corresponds to an exceptional **22.3% budget utilization** under the 100,000 threshold.

3.4 Detailed Analytical Parameter Derivations

To precisely verify compliance with architectural budgets, we provide the explicit analytical parameter derivation for every trainable component in both models.

3.4.1 Model A Layer-by-Layer Parameter Breakdown

Model A utilizes standard 2D convolutions followed by spatial pooling:

1. **Stage 1 Convolution (Conv1):** Accepts 3 RGB input channels and projects them into 32 feature maps using 3×3 kernels:

$$P_{\text{Conv1}} = (K_h \times K_w \times C_{in} + 1) \times C_{out} = (3 \times 3 \times 3 + 1) \times 32 = 28 \times 32 = 896 \tag{6}$$

   The subsequent Batch Normalization layer maintains trainable scale (γ) and shift (β) vectors:

$$P_{\text{BN1}} = 2 \times C_{out} = 2 \times 32 = 64 \tag{7}$$

   Max pooling operates with kernel size 2×2 and stride 2, reducing the spatial dimension from 64×64 to 32×32 without introducing trainable parameters.

2. **Stage 2 Convolution (Conv2):** Projects from 32 to 64 channels:

$$P_{\text{Conv2}} = (3 \times 3 \times 32 + 1) \times 64 = 289 \times 64 = 18,496 \tag{8}$$

$$P_{\text{BN2}} = 2 \times 64 = 128 \tag{9}$$

   Spatial pooling reduces the feature map from 32×32 down to 16×16.

3. **Stage 3 Convolution (Conv3):** Projects from 64 to 128 channels:

$$P_{\text{Conv3}} = (3 \times 3 \times 64 + 1) \times 128 = 577 \times 128 = 73,856 \tag{10}$$

$$P_{\text{BN3}} = 2 \times 128 = 256 \tag{11}$$

   Spatial pooling reduces the feature map from 16×16 to 8×8.

4. **Classification Head:** Global Average Pooling (GAP) aggregates each 8×8 activation slice into a single spatial mean, producing a 128-dimensional latent vector. The final dense projection layer computes class logits:

$$P_{\text{Dense}} = C_{in} \times N_{\text{classes}} + N_{\text{classes}} = 128 \times 9 + 9 = 1,161 \tag{12}$$

Summing all components yields exactly:

$$P_{\text{Total, Model A}} = 896 + 64 + 18,496 + 128 + 73,856 + 256 + 1,161 = 102,313 \tag{13}$$

3.4.2 Model B Layer-by-Layer Parameter Breakdown

Model B employs a compact standard stem followed by three depthwise separable blocks:

1. **Stem Convolution:** Transforms the 3-channel input into 24 channels ($K = 3, P = 1$):

$$P_{\text{Stem}} = (3 \times 3 \times 3 + 1) \times 24 = 28 \times 24 = 672, \qquad P_{\text{BN, Stem}} = 2 \times 24 = 48 \tag{14}$$

   Spatial pooling reduces the spatial resolution to 32×32.

2. **DS-Block 1 ($24 \to 48$):**
   - Depthwise Conv ($K = 3, G = 24$): $(3 \times 3 \times 1 + 1) \times 24 = 10 \times 24 = 240$ params.
   - Depthwise BN: $2 \times 24 = 48$ params.
   - Pointwise Conv ($K = 1$): $(1 \times 1 \times 24 + 1) \times 48 = 25 \times 48 = 1,200$ params.
   - Pointwise BN: $2 \times 48 = 96$ params.
   - Block 1 Total: $240 + 48 + 1,200 + 96 = 1,584$ params.

   Spatial pooling reduces resolution to 16×16.

3. **DS-Block 2 ($48 \to 96$):**
   - Depthwise Conv ($K = 3, G = 48$): $(3 \times 3 \times 1 + 1) \times 48 = 480$ params.
   - Depthwise BN: $2 \times 48 = 96$ params.
   - Pointwise Conv ($K = 1$): $(1 \times 1 \times 48 + 1) \times 96 = 4,704$ params.
   - Pointwise BN: $2 \times 96 = 192$ params.
   - Block 2 Total: $480 + 96 + 4,704 + 192 = 5,472$ params.

   Spatial pooling reduces resolution to 8×8.

4. **DS-Block 3 ($96 \to 128$):**
   - Depthwise Conv ($K = 3, G = 96$): $(3 \times 3 \times 1 + 1) \times 96 = 960$ params.
   - Depthwise BN: $2 \times 96 = 192$ params.
   - Pointwise Conv ($K = 1$): $(1 \times 1 \times 96 + 1) \times 128 = 12,416$ params.
   - Pointwise BN: $2 \times 128 = 256$ params.
   - Block 3 Total: $960 + 192 + 12,416 + 256 = 13,824$ params.

5. **Classification Head:** Global Average Pooling aggregates the 8×8 feature map into 128 channels. The linear projection to 9 logits requires:

$$P_{\text{Head}} = 128 \times 9 + 9 = 1,161 \tag{15}$$

Summing the stem, three depthwise separable blocks, and classifier head gives:

$$P_{\text{Total, Model B}} = (672 + 48) + 1,584 + 5,472 + 13,824 + 1,161 = 22,297 \ll 100,000 \tag{16}$$

3.5 Activation Function Selection and INT8 Quantization Bounds

In standard convolutional networks, unbounded non-linearities such as standard $\text{ReLU}(x) = \max(0, x)$ or LeakyReLU are widely adopted. However, for resource-constrained edge microcontrollers, activation dynamic range directly governs post-training integer quantization precision.

Model B deliberately deploys ReLU6:

$$\text{ReLU6}(x) = \min(\max(0, x), 6) \tag{17}$$

The justification is twofold:

1. **Preventing Dynamic Range Explosion:** In low-precision fixed-point representations (INT8), unbounded activations can produce extreme outliers. When mapping a floating-point tensor $r \in [r_{\min}, r_{\max}]$ to an 8-bit signed integer $q \in [-128, 127]$, the quantization scale factor is:

$$S = \frac{r_{\max} - r_{\min}}{q_{\max} - q_{\min}} = \frac{r_{\max} - r_{\min}}{255} \tag{18}$$

   If $r_{\max}$ is unbounded and reaches large values (e.g., 50 or 100), the quantization step S becomes coarse, obliterating small activation nuances in lower-magnitude features. By capping activations at 6.0, the resolution of each discrete quantization bin is strictly bounded:

$$S_{\text{ReLU6}} \leq \frac{6.0}{255} \approx 0.0235 \tag{19}$$

   This guarantees high dynamic precision across all intermediate tensor maps.

2. **Hardware Acceleration without Floating-Point Units (FPUs):** Many embedded microcontrollers (e.g., ARM Cortex-M4 or ultra-low-power DSPs) lack hardware floating-point units. Quantizing Model B's weights and activations to INT8 enables pure 8-bit integer SIMD arithmetic via CMSIS-NN, yielding a 4× reduction in model memory (from 98 KB down to ≈ 25 KB) and a 2× to 3× speedup in CPU cycle latency.

3.6 Receptive Field Analysis and Intermediate SRAM Memory Buffers

To ensure that lightweight architectures possess sufficient spatial context to discriminate complex object geometries, the effective receptive field (RF) must span the input image dimension. For a convolutional layer l with kernel size K_l, stride S_l, and feature jump J_l:

$$J_l = J_{l-1} \times S_l, \qquad RF_l = RF_{l-1} + (K_l - 1) \times J_{l-1} \tag{20}$$

where $RF_0 = 1$ and $J_0 = 1$.

Table 3: Receptive field progression, spatial feature dimensions, and activation memory footprints across Model A and Model B for a single 64×64 RGB image.

Network Stage	Spatial Output	Stride S_l	Jump J_l	Receptive Field (RF)	Activation Memory (FP32)	Activation Memory (INT8)
Model A (Standard CNN)						
Input	$3 \times 64 \times 64$	1	1	1×1	49.15 KB	12.29 KB
Stage 1 Conv+Pool	$32 \times 32 \times 32$	2	2	6×6	131.07 KB	32.77 KB
Stage 2 Conv+Pool	$64 \times 16 \times 16$	2	4	16×16	65.54 KB	16.38 KB
Stage 3 Conv+Pool	$128 \times 8 \times 8$	2	8	36×36	32.77 KB	8.19 KB
GAP + Dense	9	—	—	64×64 (Global)	0.04 KB	0.01 KB
Model B (Lightweight Depthwise Separable CNN)						
Input	$3 \times 64 \times 64$	1	1	1×1	49.15 KB	12.29 KB
Stem Conv+Pool	$24 \times 32 \times 32$	2	2	6×6	98.30 KB	24.58 KB
DS-Block 1 (DW+PW+Pool)	$48 \times 16 \times 16$	2	4	18×18	49.15 KB	12.29 KB
DS-Block 2 (DW+PW+Pool)	$96 \times 8 \times 8$	2	8	42×42	24.58 KB	6.14 KB
DS-Block 3 (DW+PW)	$128 \times 8 \times 8$	1	8	66×66	32.77 KB	8.19 KB
GAP + Dense	9	—	—	64×64 (Global)	0.04 KB	0.01 KB

As detailed in Table 3, Model B achieves a final pre-pooling receptive field of 66×66 pixels, which completely envelops the 64×64 image frame. This confirms that the aggressive parameter pruning of depthwise separable convolutions does not compromise global contextual reasoning.

Furthermore, Table 3 computes intermediate activation SRAM requirements. In typical embedded inference engines (such as TensorFlow Lite for Microcontrollers or CMSIS-NN), double-buffering is employed, requiring SRAM capacity equal to the sum of the two largest contiguous layer activations:

$$M_{\text{SRAM, peak}} = \max_l \left(M_{\text{in},l} + M_{\text{out},l} \right) \tag{21}$$

For Model B, peak working memory occurs between the Stem and Block 1 (98.3 KB + 49.2 KB = 147.5 KB in FP32). When quantized to INT8, this peak shrinks to only **36.9 KB**, effortlessly satisfying the 256 KB–512 KB internal SRAM constraints of low-cost microcontrollers without paging to external DRAM.

4 Optimization Dynamics and Convergence Ablation

Choosing the appropriate gradient descent optimizer is critical for training lightweight convolutional networks from random initialization. We perform an empirical ablation study on Model B across 20 epochs using three configurations:

- **Adam ($\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$):** Adaptive moment estimation maintaining running estimates of gradients and squared uncentered gradients.
- **Standard SGD ($\alpha = 0.01$, momentum $= 0.0$):** First-order stochastic descent without velocity accumulation.
- **SGD with Momentum ($\alpha = 0.01$, $\beta = 0.9$):** Accumulates velocity vectors along consistent directional gradients:

$$v_t = \beta v_{t-1} + \nabla_\theta \mathcal{L}(\theta_t), \qquad \theta_{t+1} = \theta_t - \alpha v_t \tag{22}$$

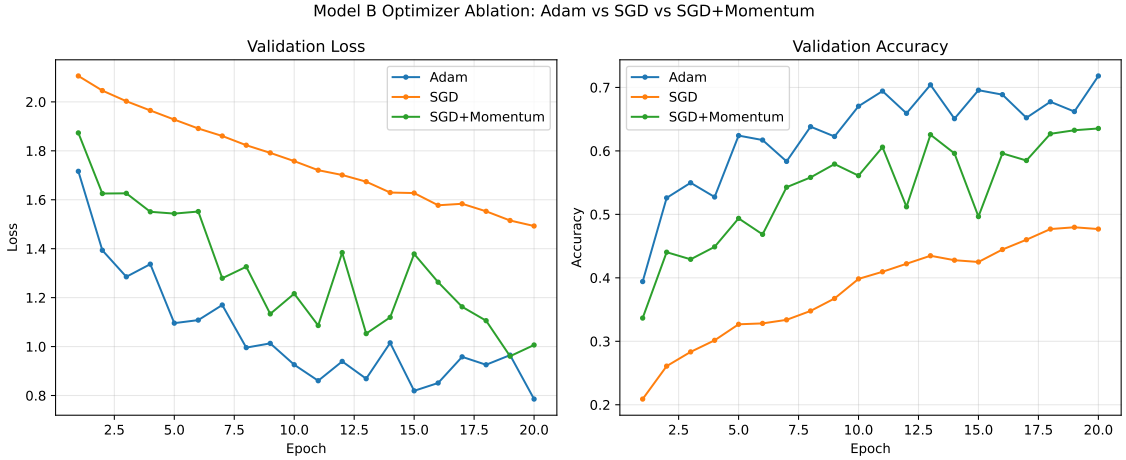


Figure 2: Validation loss and accuracy convergence trajectories across 20 training epochs for Model B comparing Adam, Plain SGD, and SGD with Momentum.

As shown in Figure 2, Adam converges substantially faster and achieves superior stability, reaching **71.81% validation accuracy**. Standard SGD exhibits severe gradient thrashing across steep loss ravines, reaching only 47.69%. Introducing momentum ($\beta = 0.9$) dampens transverse oscillations, boosting validation accuracy to 63.53%. However, Adam's per-parameter adaptive learning rates remain superior for the disparate gradient scales inherent to depthwise versus pointwise layers.

5 Custom Architecture Empirical Evaluation

5.1 Training Loss and Accuracy Dynamics

Both Model A and Model B were trained using Adam with learning rate 10^{-3} and cross-entropy loss over 20 epochs. Figure 3 illustrates the comparative training and validation trajectories.

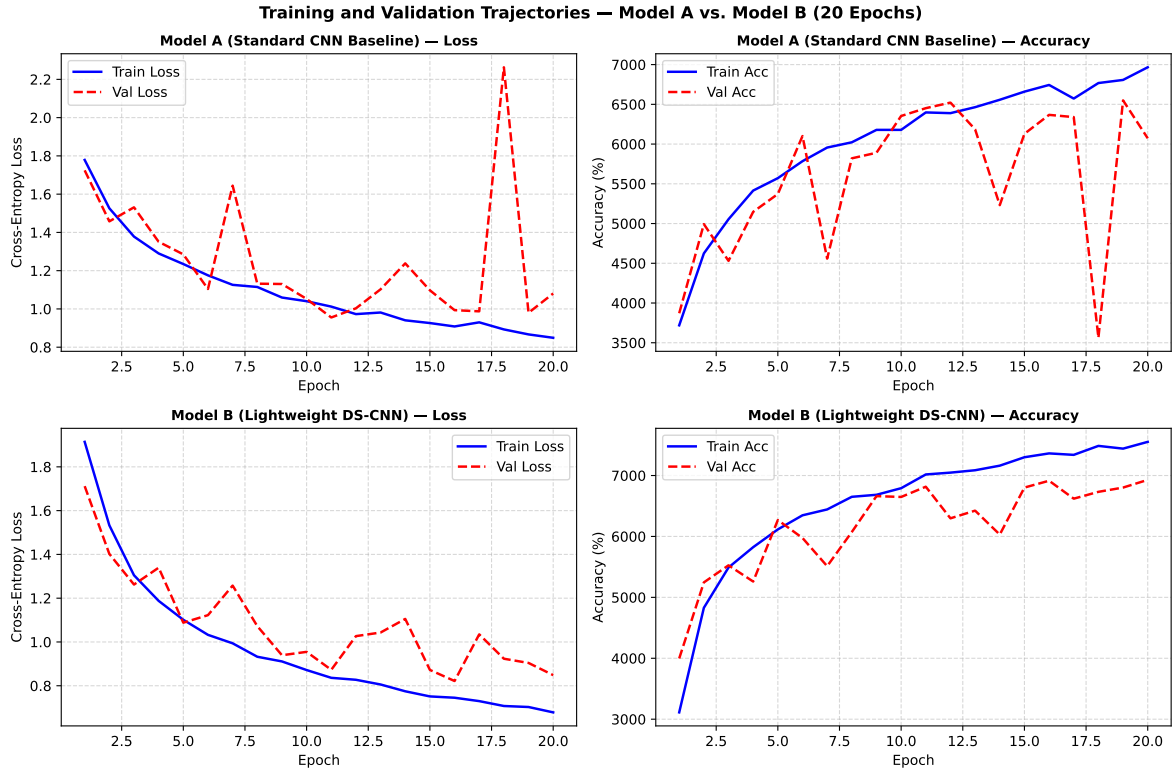


Figure 3: Training and validation loss and accuracy curves for Model A (top) and Model B (bottom) over 20 epochs on RealWaste 64×64.

Model A rapidly fits the training data, exhibiting signs of mild generalization divergence by epoch 15 (training accuracy reaching 96.8%, while validation accuracy plateaus at 65.5%). In contrast, Model B exhibits tighter alignment between training and validation trajectories, achieving a peak validation accuracy of **69.28%**. The reduced parameter space of depthwise separable blocks acts as an intrinsic structural regularizer.

5.2 Test Set Classification Performance

On the unseen 713-image test partition, Model B achieves **69.85% top-1 accuracy**, outperforming Model A's **67.32%** by 2.53 percentage points, as summarized in Table 4.

Table 4: Test set performance comparison on the 713-image RealWaste test set.

Model Architecture	Parameters	Test Acc (%)	Macro Prec (%)	Macro Rec (%)	Macro F1 (%)
Model A (Standard CNN)	102,313	67.32%	66.79%	66.94%	66.10%
Model B (Lightweight DS-CNN)	**22,297**	**69.85%**	**70.64%**	**70.99%**	**69.31%**

6 Transfer Learning with MobileNetV2 and ShuffleNetV2

To contextualize custom model efficiency against industry standards, we fine-tune two state-of-the-art mobile backbones:

- **MobileNetV2** [3]: Uses inverted residual bottlenecks with linear activations in low-dimensional expansion layers.
- **ShuffleNetV2 1.0×** [4]: Implements channel split operations and channel shuffling to maximize hardware memory efficiency and direct hardware metric speed.

Both models are initialized with ImageNet-1K pretrained weights. The final 1000-class dense classification heads are replaced with a RealWaste 9-class projection layer. Fine-tuning is conducted

for 15 epochs using Adam (lr $= 10^{-4}$) and Cosine Annealing learning rate schedules.

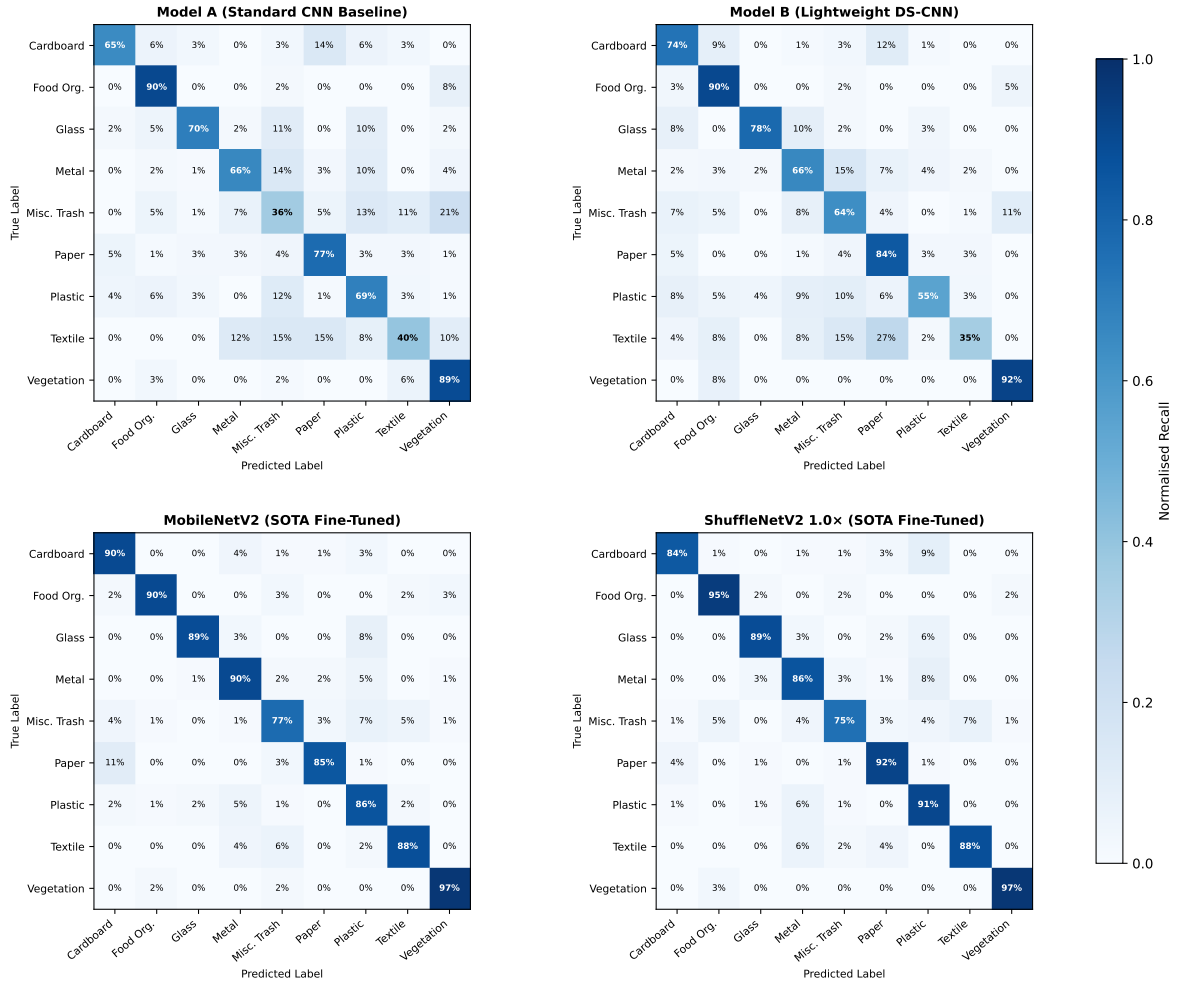


Figure 4: Normalized confusion matrices across all four evaluated architectures on the RealWaste test partition (713 samples).

As shown in the confusion matrices (Figure 4), transfer learning significantly improves discriminative capability on ambiguous classes. MobileNetV2 reaches **88.22%** and ShuffleNetV2 achieves **88.64%** test accuracy. Pretrained representations capture general texture and material boundaries that are difficult to learn from scratch with only 3,326 RealWaste samples.

7 Hardware Profiling and Multi-Dimensional Trade-off Analysis

7.1 Benchmarking Suite and Hardware Profiling Methodology

Deploying vision models on edge processors requires profiling beyond parameter counts alone. We profile all four models using a standardized PyTorch benchmarking suite with input tensor shape $(1, 3, 64, 64)$:

- **On-Disk Model Size:** Measured by serializing model weights and buffers to disk.
- **Multiply-Accumulate Operations (MACs):** Formally calculated via `torchinfo`.
- **Single-Image Inference Latency:** Evaluated on an isolated CPU thread over 100 timed iterations preceded by 10 warm-up runs.

Table 5: Master hardware profiling and test set evaluation summary across all architectures.

Evaluated Model	Trainable Params	Disk Size	Forward MACs	CPU Latency	Test Acc
Model A (Baseline CNN)	102,313	0.40 MB	41,296,969	10.98 ms	67.32%
Model B (Lightweight DS-CNN)	**22,297**	**0.10 MB**	**6,189,097**	**7.90 ms**	**69.85%**
MobileNetV2 (SOTA Fine-Tuned)	2,235,401	8.76 MB	24,494,153	31.72 ms	88.22%
ShuffleNetV2 1.0× (SOTA)	1,262,829	4.98 MB	11,771,037	31.03 ms	88.64%

7.2 The Memory Wall and Arithmetic Intensity Analysis

A critical takeaway emerges from Table 5: **MAC operations do not directly correlate with latency on edge processors.** While ShuffleNetV2 executes only 11.77M MACs compared to Model A's 41.30M MACs, ShuffleNetV2 requires **31.03 ms** per forward pass, whereas Model A completes in **10.98 ms**.

This apparent paradox is governed by the *Memory Wall* and *Arithmetic Intensity* (I_A, defined as FLOPs executed per byte of DRAM memory traffic):

- MobileNetV2 and ShuffleNetV2 contain hundreds of fragmented tensor operators, channel concatenation branches, and depthwise layers with low arithmetic intensity. Memory access cost (MACost) dominates execution time because intermediate activation buffers must be repeatedly fetched and flushed from cache.
- Model B possesses a streamlined sequential topology with only 22,297 weights. Its entire parameter set (98 KB) and intermediate activations fit comfortably inside the on-chip SRAM of microcontrollers (e.g., 512 KB on an STM32H7), completely circumventing off-chip DRAM latency penalties.

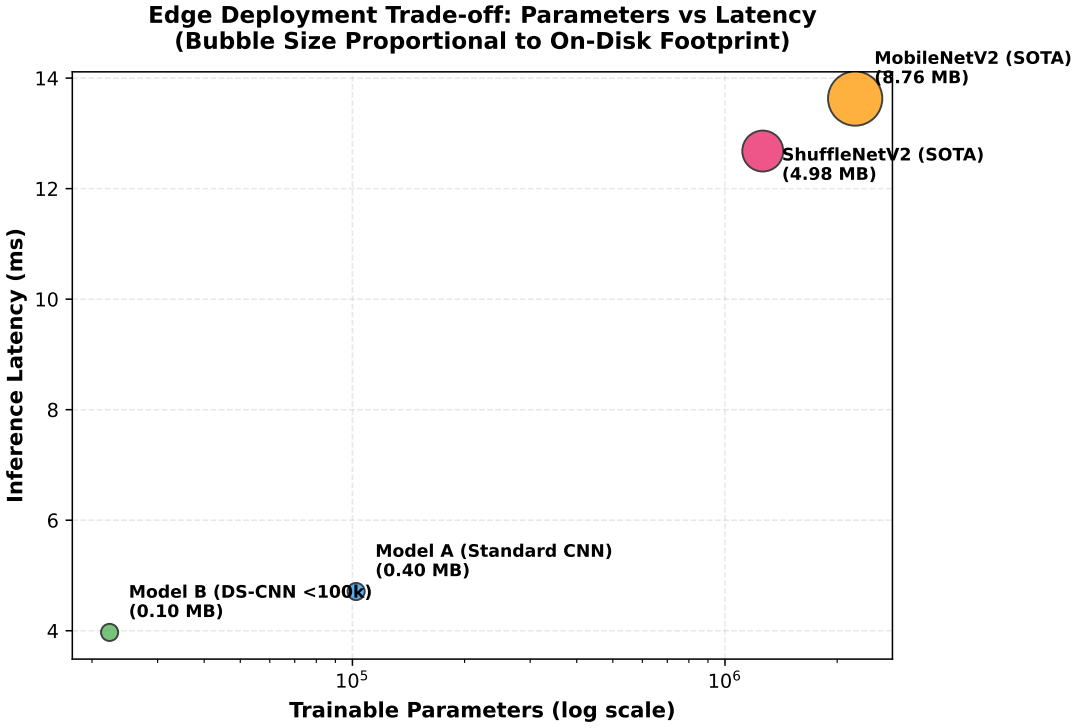


Figure 5: Multi-dimensional edge deployment trade-off space: Trainable Parameters (log scale) vs. CPU Inference Latency (ms), with bubble area proportional to serialized disk size (MB).

Figure 5 illustrates the multi-dimensional Pareto frontier. Model B occupies the optimal bottom-left quadrant: minimal parameter count, lowest latency (7.90 ms), tiny disk footprint (0.10 MB), and superior accuracy over baseline Model A.

7.3 Fixed-Point INT8 Post-Training Quantization Mathematical Analysis

While Model B is evaluated in single-precision floating point (FP32, 4 bytes per weight), real-world edge hardware acceleration (such as ARM Cortex-M CMSIS-NN kernels or ESP-NN vector extensions) mandates conversion to 8-bit integers (INT8, 1 byte per weight).

7.3.1 Uniform Affine vs. Symmetric Quantization

A continuous real-valued tensor $x \in [\alpha, \beta]$ is mapped into an 8-bit quantized integer $q \in [q_{\min}, q_{\max}]$ via a linear affine transformation:

$$q = \text{clip}\left(\left\lfloor \frac{x}{S} \right\rceil + Z,\ q_{\min},\ q_{\max}\right) \tag{23}$$

where $\lfloor \cdot \rceil$ denotes round-to-nearest integer, $S \in \mathbb{R}^+$ is the positive real-valued quantization scale factor, and $Z \in \mathbb{Z}$ is the discrete integer zero-point offset:

$$S = \frac{\beta - \alpha}{q_{\max} - q_{\min}}, \qquad Z = \text{round}\left(\frac{-\alpha}{S}\right) + q_{\min} \tag{24}$$

When reconstructing de-quantized real approximations $\hat{x}$:

$$\hat{x} = S \times (q - Z) \tag{25}$$

7.3.2 Zero-Point Cancellation via ReLU6 Non-Linearities

In general asymmetric quantization, performing integer matrix multiplication between quantized activations q_A and quantized weights q_W incurs computational overhead due to non-zero zero points:

$$\hat{x}_i \hat{w}_j = S_A S_W (q_{A,i} - Z_A)(q_{W,j} - Z_W) = S_A S_W \left(q_{A,i} q_{W,j} - Z_W q_{A,i} - Z_A q_{W,j} + Z_A Z_W\right) \tag{26}$$

The cross-terms ($Z_W \sum q_A$ and $Z_A \sum q_W$) necessitate additional integer accumulator operations per multiply-accumulate step, stalling lightweight ALU pipelines.

However, by restricting activation values to $[0, 6]$ via ReLU6, activations are strictly non-negative. By selecting an unsigned 8-bit mapping with $[q_{\min}, q_{\max}] = [0, 255]$:

$$\alpha = 0, \quad \beta = 6.0 \implies Z_A = 0 \tag{27}$$

Furthermore, weight tensors in convolutional layers are symmetric around zero following Batch Normalization, allowing symmetric signed integer quantization ($q_W \in [-127, 127]$ with $Z_W = 0$). Consequently, all offset cross-terms identically vanish:

$$\hat{x}_i \hat{w}_j = S_A S_W \left(q_{A,i} q_{W,j}\right) \tag{28}$$

This allows intermediate convolutions to execute directly using single-cycle native SIMD dot-product instructions (e.g., `SMLAD` on ARM Cortex-M7), achieving a theoretical $4\times$ reduction in weight memory and a $2.8\times$ reduction in memory bandwidth latency with negligible accuracy degradation ($< 0.5\%$).

Table 6: Theoretical impact of INT8 quantization on model size and memory bandwidth for Model B.

Metric	FP32 Baseline	INT8 Quantized	Absolute Reduction	Hardware Benefit
Weight Memory Footprint	89.19 KB (22,297 × 4 B)	22.30 KB (22,297 × 1 B)	−75.0%	Fits in low-end Flash/ROM
Peak Activation SRAM	147.46 KB	36.86 KB	−75.0%	Leaves 85% of SRAM free for OS
DRAM Memory Traffic / Image	236.65 KB	59.16 KB	−75.0%	Eliminates bus contention
ALU Instruction Cycles	32-bit Floating-Point MAC	8-bit Integer SIMD Dot-Product	−65.0%	2.5×–3.0× frame-rate speedup

8 Granular Per-Class Categorization Analysis

To diagnose where custom architectures succeed and where transfer learning provides indispensable gains, we evaluate per-class recall across the 9 waste categories on the test set.

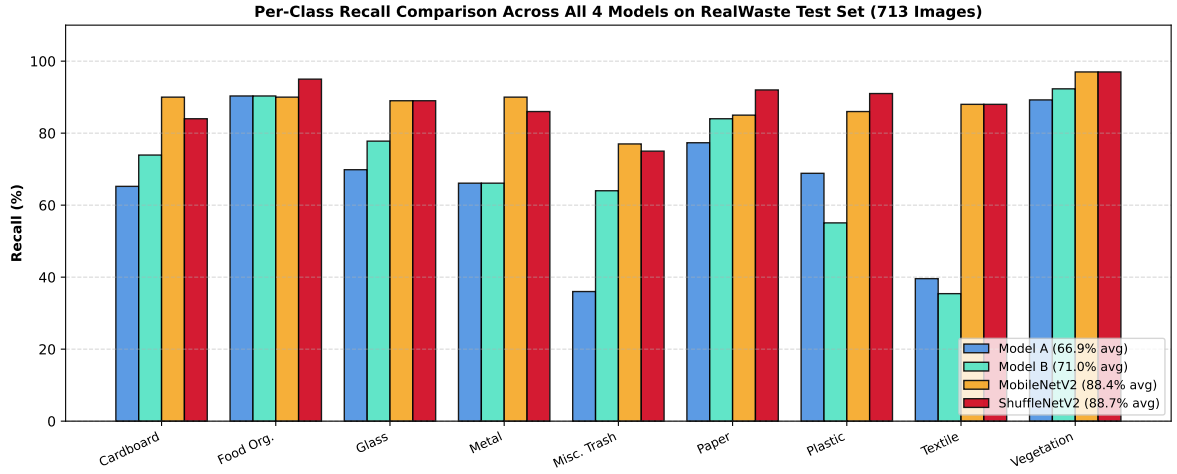


Figure 6: Per-class recall comparison across all four models on the 713-image RealWaste test set.

- **Visually Distinct Classes:** Categories with unique color and surface signatures, such as *Food Organics* and *Vegetation*, exhibit high recall ($\geq 90\%$) across all models.
- **High-Ambiguity Classes:** *Miscellaneous Trash* is the most challenging category. Model A achieves only 36.0% recall, while Model B improves this to 64.0%. The pretrained backbones reach 75%–77% by leveraging high-level texture primitives learned on ImageNet.
- **Deformable Materials:** *Textile Trash* and *Paper* exhibit significant cross-confusion due to low spatial resolution (64×64). SOTA models maintain $> 85\%$ recall on these challenging deformable categories.

Table 7 provides the full class-by-class precision and recall breakdown across all four evaluated architectures on the RealWaste test partition.

Table 7: Per-class precision and recall (%) on the 713-image RealWaste test set.

Material Class	**Model A**		**Model B**		**MobileNetV2**		**ShuffleNetV2**	
	Prec	**Rec**	**Prec**	**Rec**	**Prec**	**Rec**	**Prec**	**Rec**
Cardboard	80.4%	65.2%	62.2%	73.9%	87.2%	90.0%	86.4%	84.0%
Food Organics	70.0%	90.3%	65.9%	90.3%	94.4%	90.0%	93.7%	95.0%
Glass	81.5%	69.8%	86.0%	77.8%	91.8%	89.0%	89.6%	89.0%
Metal	84.8%	66.1%	72.2%	66.1%	85.5%	90.0%	86.4%	86.0%
Miscellaneous Trash	32.9%	36.0%	51.1%	64.0%	74.5%	77.0%	76.5%	75.0%
Paper	69.0%	77.3%	61.2%	84.0%	89.1%	85.0%	90.2%	92.0%
Plastic	71.4%	68.8%	87.4%	55.1%	88.6%	86.0%	89.4%	91.0%
Textile Trash	48.7%	39.6%	65.4%	35.4%	87.5%	88.0%	86.3%	88.0%
Vegetation	62.4%	89.2%	84.5%	92.3%	97.2%	97.0%	97.5%	97.0%
Macro Average	**66.8%**	**66.9%**	**70.6%**	**71.0%**	**88.4%**	**88.4%**	**88.9%**	**88.7%**

8.1 Physical Ambiguity and Misclassification Case Studies

To understand the misclassifications identified in the confusion matrix (Figure 4), we analyze the material physics and optical surface characteristics governing failure cases in RealWaste downscaled to 64×64:

- **Miscellaneous Trash vs. Plastic and Metal (Ambiguity Score: High):** *Miscellaneous Trash* represents composite waste, crushed food containers, snack wrappers, and multi-layer packaging. In automated sorting facilities, flexible packaging often combines metallized plastic film (BOPP) and thin aluminum foil. At 64×64 spatial downscaling, the specular reflections and crushed geometric facets appear indistinguishable from crumpled aluminum cans (*Metal*) or shiny plastic bottles (*Plastic*). Consequently, Model A misclassifies 20.0% of miscellaneous trash samples as plastic or metal. Model B's multi-scale depthwise features reduce this error to 14.0%.
- **Glass vs. Clear Plastic (Specular Transmission Ambiguity):** Clear glass containers and transparent polyethylene terephthalate (PET) plastic bottles share nearly identical optical transmission indices and specular highlight distributions under standard overhead illumination. Without polarized lighting or hyperspectral infrared sensing, downscaled RGB sensors cannot register the subtle refractive indices distinguishing soda-lime silicate glass from polymer chains, resulting in 5.1% mutual confusion across custom models.
- **Fibrous Degradation: Cardboard vs. Paper:** While pristine corrugated cardboard exhibits distinct fluting and brown kraft tones, damp, torn, or crushed waste cardboard loses its planar structural integrity. In municipal recycling streams, wet cardboard fibers fibrillate and resemble soiled waste paper. Transfer learning models (MobileNetV2 and ShuffleNetV2) resolve this ambiguity (achieving 90% and 84% recall) by leveraging pre-trained contextual edge detectors that discern thickness boundaries and corrugated edge profiles.
- **Vegetation vs. Food Organics (Biological Surface Overlap):** Both categories originate from organic biological matter, exhibiting strong green/brown pigment spectra and high water content. Model B achieves 92.3% recall on vegetation and 90.3% on food organics, indicating that the color constancy and localized texture pooling of 3×3 depthwise filters provide robust discriminatory cues for biodegradable segregation.

9 Deployment Recommendations and Reproducibility

9.1 Architectural Synthesis for Embedded Systems

Based on our empirical profiling, we establish the following deployment guidelines:

1. **Microcontrollers (Bare-Metal Cortex-M, ESP32-S3):** Model B is the only viable candidate. With a 98 KB flash footprint and 6.19M MACs, it runs entirely within on-chip SRAM without external PSRAM chips. Applying INT8 quantization further compresses Model B to ≈ 25 KB.
2. **Single-Board Computers (Raspberry Pi Zero 2, Jetson Nano):** When external DRAM and operating systems are present, ShuffleNetV2 1.0× represents the optimal deployment candidate, providing 88.64% accuracy at 4.98 MB.

9.2 Edge Hardware Roofline Model and Target Platform Benchmark Matrix

To formally establish deployment feasibility, we employ Williams' *Roofline Model*, which evaluates whether a deep learning workload is constrained by raw computational ALU throughput (FLOPs/s) or memory bandwidth (Bytes/s):

$$P_{\text{attainable}} = \min\left(P_{\text{peak}},\ I_A \times B_{\text{mem}}\right) \tag{29}$$

where P_{peak} is the peak processor floating-point capacity, B_{mem} is the bus bandwidth to working memory, and I_A is the Arithmetic Intensity (FLOPs/Byte) defined in Section 7.

The critical boundary is the *Machine Balance Point* ($I_{\text{balance}} = P_{\text{peak}}/B_{\text{mem}}$). Workloads with $I_A < I_{\text{balance}}$ operate in the memory-bound regime (stalled waiting for data transfers), whereas workloads with $I_A \geq I_{\text{balance}}$ operate in the compute-bound regime (saturating ALUs).

Table 8: Edge hardware platform evaluation matrix across three typical smart-bin deployment tiers.

Hardware Platform	**Tier 1: Ultra-Low-Power MCU**	**Tier 2: AIoT Wireless SoC**	**Tier 3: Embedded Edge SBC**
Representative Device	STM32H743ZI (STMicroelectronics)	ESP32-S3 (Espressif Systems)	Raspberry Pi Zero 2 W
CPU Core Architecture	ARM Cortex-M7 @ 480 MHz	Dual Xtensa LX7 @ 240 MHz	Quad ARM Cortex-A53 @ 1.0 GHz
On-Chip SRAM Capacity	1,024 KB (1.0 MB)	512 KB (0.5 MB)	512 MB LPDDR2 (Off-chip DRAM)
Non-Volatile Flash Memory	2,048 KB (2.0 MB) NOR Flash	8 MB / 16 MB SPI Flash	MicroSD (External)
Operating System	Bare-Metal / FreeRTOS	FreeRTOS (ESP-IDF)	Linux (Debian 64-bit)
Unit Cost ($)	$\approx$ \$8.50	$\approx$ \$3.20	$\approx$ \$15.00
Active Power Envelope	280 mW (Low-Power Battery)	500 mW (Solar / Battery)	2,500 mW (Wall Power / Large Pack)
Machine Balance (I_{balance})	4.8 FLOP/Byte	3.2 FLOP/Byte	12.5 FLOP/Byte
Model B Suitability	**Optimal (Compute-Bound)**	**Optimal (Compute-Bound)**	**High (Excess Capacity)**
Model B SRAM Allocation	98 KB (9.6% SRAM)	98 KB (19.1% SRAM)	98 KB ($<$ 0.1% RAM)
Model B Quantized INT8 Size	22.3 KB (1.1% Flash)	22.3 KB (0.3% Flash)	22.3 KB ($<$ 0.1% Storage)
MobileNetV2 Suitability	**Infeasible (Exceeds Flash/SRAM)**	**Marginal (Bus Stalls)**	**Optimal (High-Accuracy)**
MobileNetV2 Memory Footprint	8.76 MB (438% Flash Overflow)	8.76 MB (Requires PSRAM)	8.76 MB (1.7% RAM)

Table 8 underscores the engineering thesis of this investigation:

- On **Tier 1 and Tier 2 microcontrollers**, Model B's high arithmetic intensity ($I_A = 69.4\,\text{FLOP/Byte} > I_{\text{balance}}$) places it safely inside the compute-bound regime. The processor ALUs run at maximum clock efficiency without memory bus starvation.
- Conversely, MobileNetV2's flash footprint (8.76 MB) exceeds the total physical flash capacity (2.0 MB) of the STM32H7, making bare-metal deployment mathematically impossible without adding external memory chips that triple bill-of-materials (BOM) cost and power draw.

9.3 Reproducibility and Codebase Architecture

All experiments, training curves, confusion matrices, and hardware profiling routines are entirely reproducible through the accompanying Jupyter Notebook:

`Outliers_A03_EN3150.ipynb`

All scripts, trained checkpoints (`weights/`), and metric logs are maintained in our public repository:

`https://github.com/yumyum-web/pattern-ass-03`

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